King's College London

Evaluating LLMs & SLMs for humanities research

Judging language models on the tasks you actually do

arc:apex arc:nexus arc:lite arc:nano

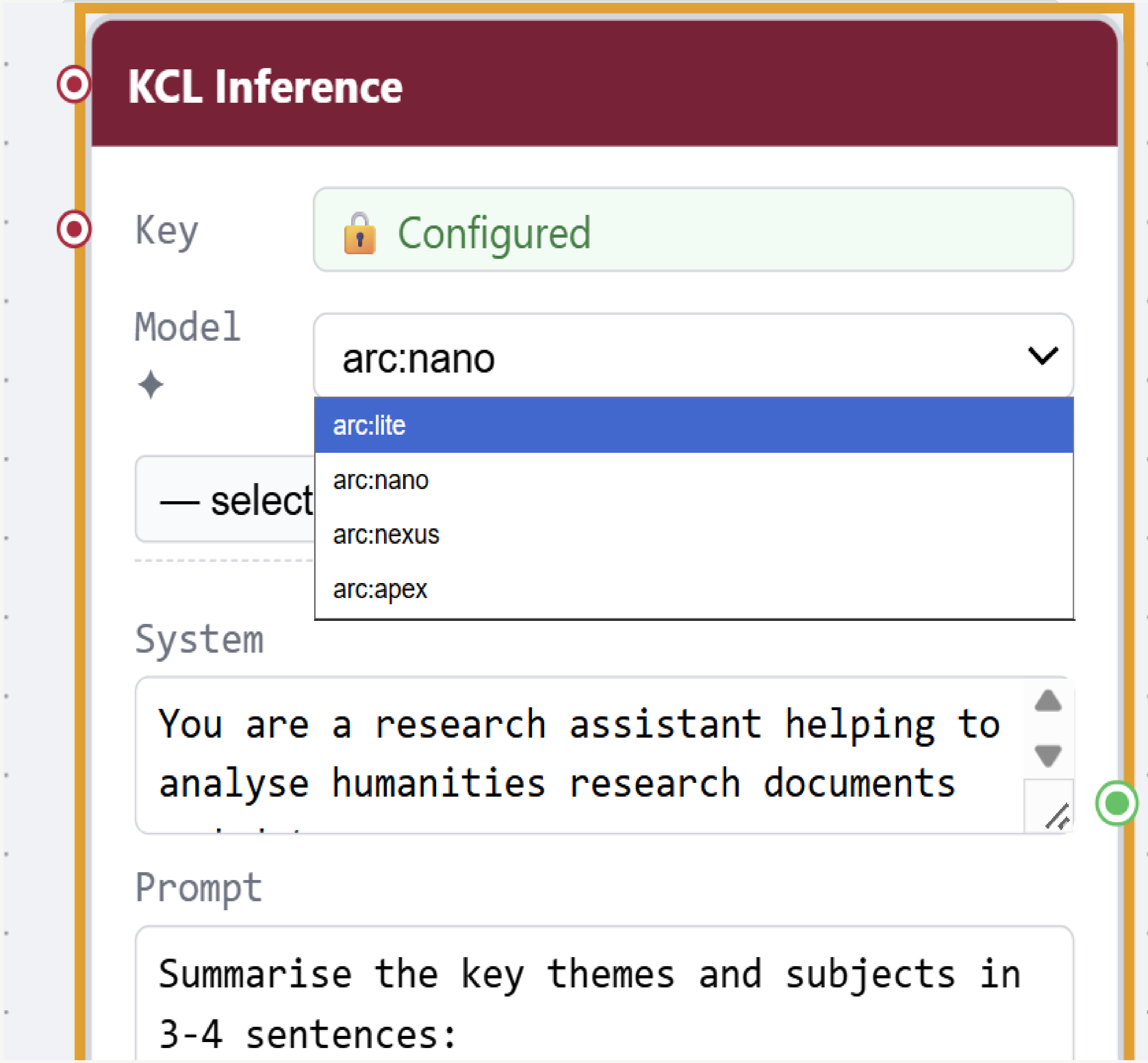
PART 0

The vocabulary

What are these things, before we judge them?

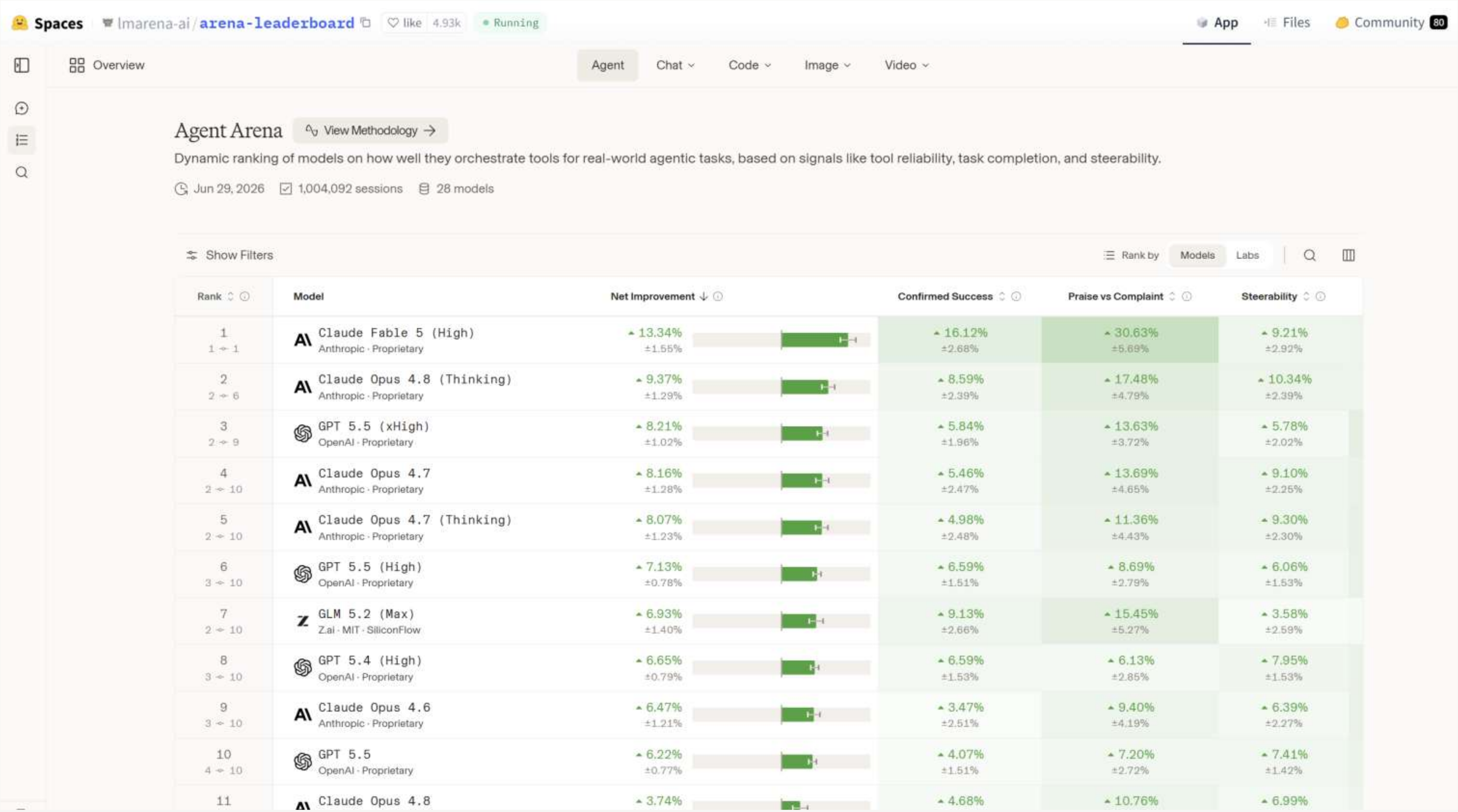
https://nfcs-poc.sites.er.kcl.ac.uk

Four names on a dropdown. What's actually behind them?



- `arc:apex` the flagship
- `arc:nexus` the workhorse
- `arc:lite` the everyday
- `arc:nano` the small one

Four names on a dropdown. What's actually behind them?



Spaces lmarena-ai/arena-leaderboard like 4.93k Running

App Files Community 80

Overview

Agent Chat Code Image Video

Agent Arena

View Methodology →

Dynamic ranking of models on how well they orchestrate tools for real-world agentic tasks, based on signals like tool reliability, task completion, and steerability.

Jun 29, 2026 1,004,092 sessions 28 models

Show Filters

Rank by Models Labs

Rank	Model	Net Improvement	Confirmed Success	Praise vs Complaint	Steerability
1 1 ← 1	Claude Fable 5 (High) Anthropic · Proprietary	▲ 13.34% ±1.55%	▲ 16.12% ±2.68%	▲ 30.63% ±5.69%	▲ 9.21% ±2.92%
2 2 ← 6	Claude Opus 4.8 (Thinking) Anthropic · Proprietary	▲ 9.37% ±1.29%	▲ 8.59% ±2.39%	▲ 17.48% ±4.79%	▲ 10.34% ±2.39%
3 2 ← 9	GPT 5.5 (xHigh) OpenAI · Proprietary	▲ 8.21% ±1.02%	▲ 5.84% ±1.96%	▲ 13.63% ±3.72%	▲ 5.78% ±2.02%
4 2 ← 10	Claude Opus 4.7 Anthropic · Proprietary	▲ 8.16% ±1.28%	▲ 5.46% ±2.47%	▲ 13.69% ±4.65%	▲ 9.10% ±2.25%
5 2 ← 10	Claude Opus 4.7 (Thinking) Anthropic · Proprietary	▲ 8.07% ±1.23%	▲ 4.98% ±2.48%	▲ 11.36% ±4.43%	▲ 9.30% ±2.30%
6 3 ← 10	GPT 5.5 (High) OpenAI · Proprietary	▲ 7.13% ±0.78%	▲ 6.59% ±1.51%	▲ 8.69% ±2.79%	▲ 6.06% ±1.53%
7 2 ← 10	GLM 5.2 (Max) Z.ai · MIT · SiliconFlow	▲ 6.93% ±1.40%	▲ 9.13% ±2.66%	▲ 15.45% ±5.27%	▲ 3.58% ±2.59%
8 3 ← 10	GPT 5.4 (High) OpenAI · Proprietary	▲ 6.65% ±0.79%	▲ 6.59% ±1.53%	▲ 6.13% ±2.85%	▲ 7.95% ±1.53%
9 3 ← 10	Claude Opus 4.6 Anthropic · Proprietary	▲ 6.47% ±1.21%	▲ 3.47% ±2.51%	▲ 9.40% ±4.19%	▲ 6.39% ±2.27%
10 4 ← 10	GPT 5.5 OpenAI · Proprietary	▲ 6.22% ±0.77%	▲ 4.07% ±1.51%	▲ 7.20% ±2.72%	▲ 7.41% ±1.42%
11	Claude Opus 4.8	▲ 3.74%	▲ 4.68%	▲ 10.76%	▲ 6.99%

https://huggingface.co/spaces/lmarena-ai/arena-leaderboard

Tokens: the unit of everything

A model reads *tokens* , not words — chunks of roughly ¾ of a word. Every limit and every cost is counted in tokens.

`The` `paleo` `graph` `er` `dated` `the` `fib` `ula` one rare word → three tokens

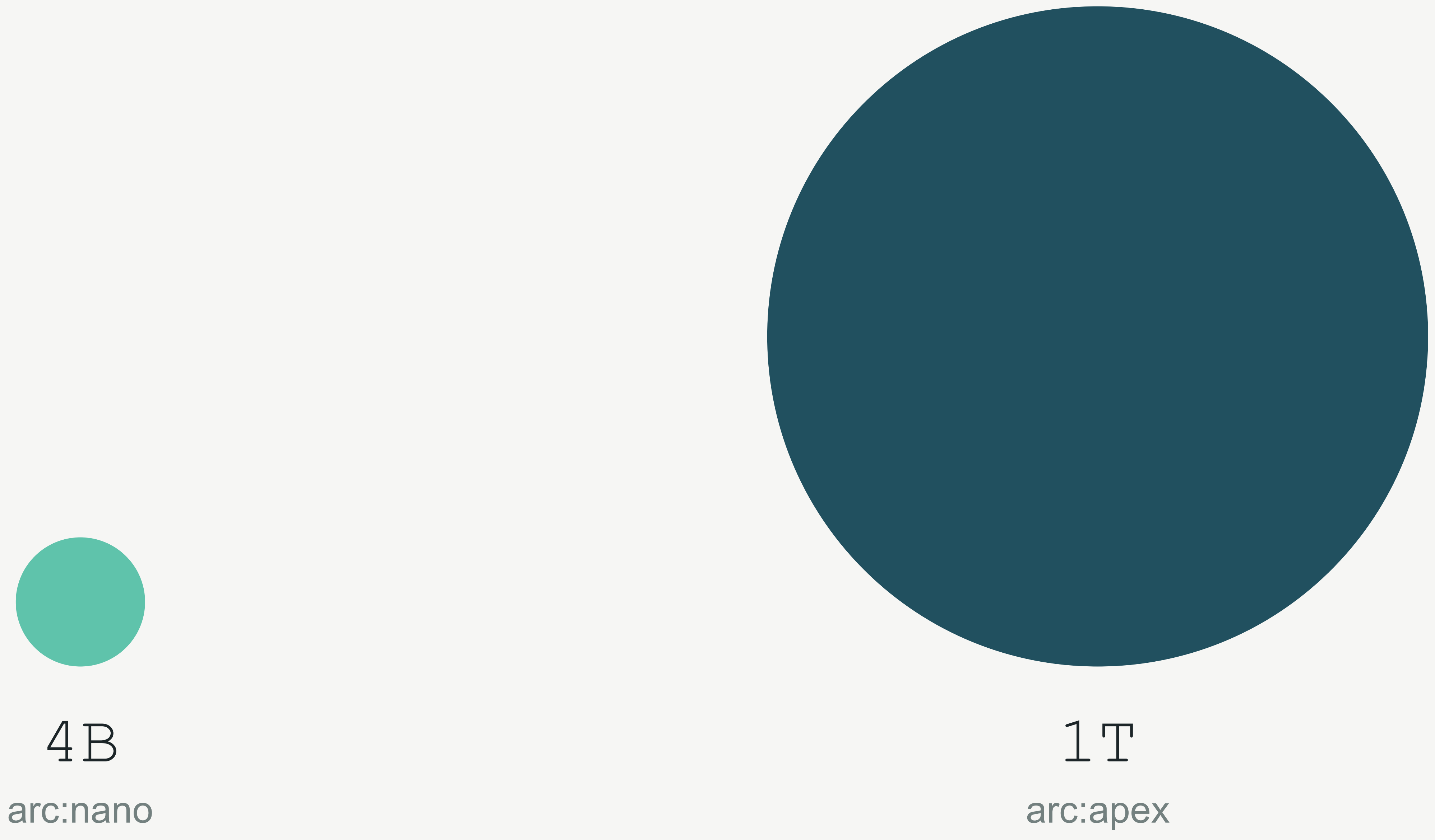
262,144 tokens
a model's context limit

≈

~190,000 words
not 262,144 words

Parameters: the size of a model

Parameters are the model's learned weights — more parameters means more room to store patterns. "Size" means this count. But size alone is misleading.



250× the parameters — but not 250× the usefulness for a fixed task.

Active vs total parameters

Mixture-of-Experts (MoE) lets total parameters balloon while only a few "experts" fire per token. Active drives cost & speed; total relates to knowledge.

	apex	nexus	lite	nano
Total	1T	35B	26B	~4B
Active	32B	~3B	~4B	4B (all)
Architecture	MoE	MoE	MoE	Dense

apex holds a trillion parameters but only fires 32B per token — that 32B is what you pay for.

Quantisation: the precision of the numbers

Weights can be stored at lower precision to save memory and run faster, with some fidelity lost. `INT4`, `NVFP4`, `Q4_K_M` all mean roughly 4-bit — not capability tiers, the same model compressed.

16-BIT · FULL PRECISION		4-BIT · QUANTISED
0.38172649	→	0.375
faithful · large · slower		approximate · small · faster

Context window: what fits

Model	Context	≈ words
apex / nexus	262,144	~190,000 — a monograph
lite	32,768	~24,000 — a long article
nano	8,192	~6,000 — a journal article

The conversation grows

The model is stateless — the whole history is replayed every turn.

1 Cost grows with the *square* of conversation length.

2 When the window fills, whole early chunks drop — no gradual fade.

turn 1 — dropped

turn 2

turn 3

turn 4

turn 5 — all of the above, resent

Lost in the middle

Models recall the *start* and *end* of a long context more reliably than the middle. A big window means the data fits — not that every part is read equally well.

start · reliable　　middle · faded　　end · reliable

Flavours: base, instruct, reasoning — and how small is "small"?

`base`

continues text

`instruct`

follows instructions — all ARC models

`reasoning`

"thinks" before answering

"SLM" band — small ≠ useless

`nano` `lite` `nexus` `apex`

SMALL — LARGE

PART 1

Reading the evidence

What do we actually know when a model scores 87%?

87%?

What do we actually know when we know a model scores 87% on MMLU?

A benchmark is a proxy

A structured test standing in for a capability we care about — because measuring the real thing directly is too slow, costly or subjective. The gap between proxy and reality is where the trouble lives.

A fixed reference point — useful for relative measurement, not absolute truth.

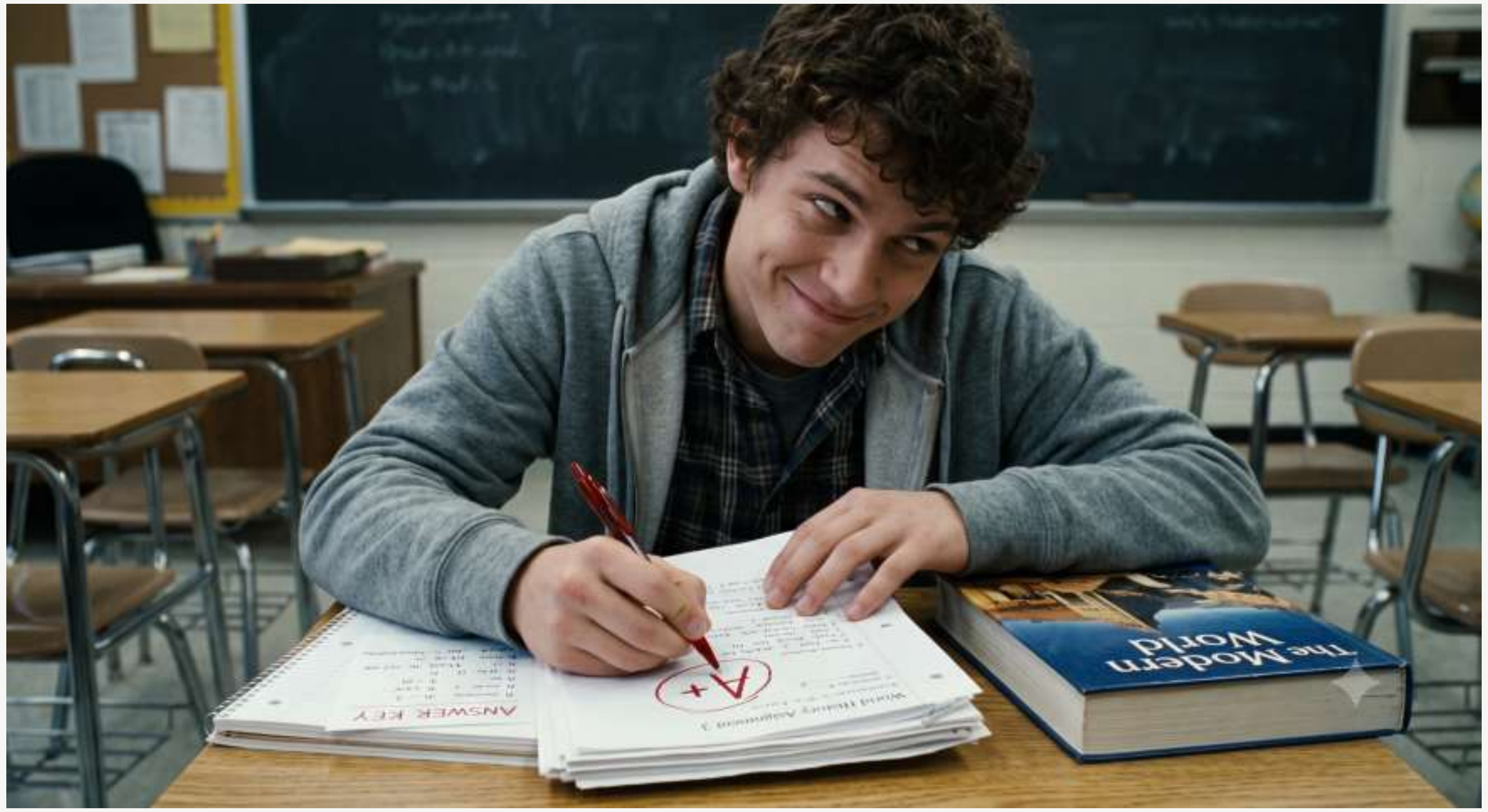
The flavours of benchmark

Benchmark	Format	Tests	Doesn't test
MMLU	Multiple choice, 57 subjects	Breadth of factual knowledge	Open generation, reasoning quality
MATH / GSM8K	Short-answer maths	Mathematical reasoning	Verbal reasoning, ambiguity
HumanEval / MBPP	Code generation	Programming	Almost everything else
HellaSwag / ARC	Multiple choice	Commonsense plausibility	Multi-step tasks
TruthfulQA	MC / generation	Resisting falsehoods	Task performance
LMSYS Arena	Human preference votes	Subjective satisfaction	Specific task accuracy

The "doesn't test" column is always long. **None of these is *your* task.**

The contamination problem

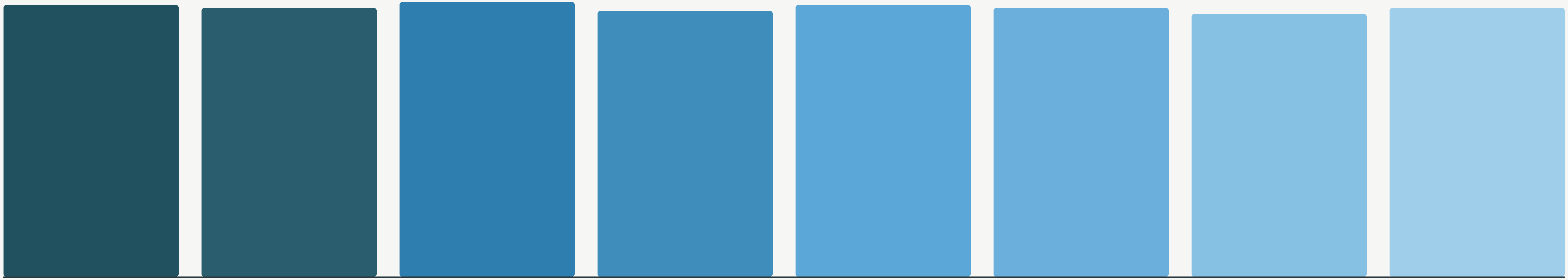
If benchmark questions appeared in training data, the model *recalls* rather than reasons. Training sets are vast and unaudited — so contamination is the default assumption.



> High scores on popular public benchmarks deserve more scepticism than scores on fresh or private tests.

The saturation problem

When top models cluster near the ceiling, the benchmark can't tell them apart — the ranking becomes noise.

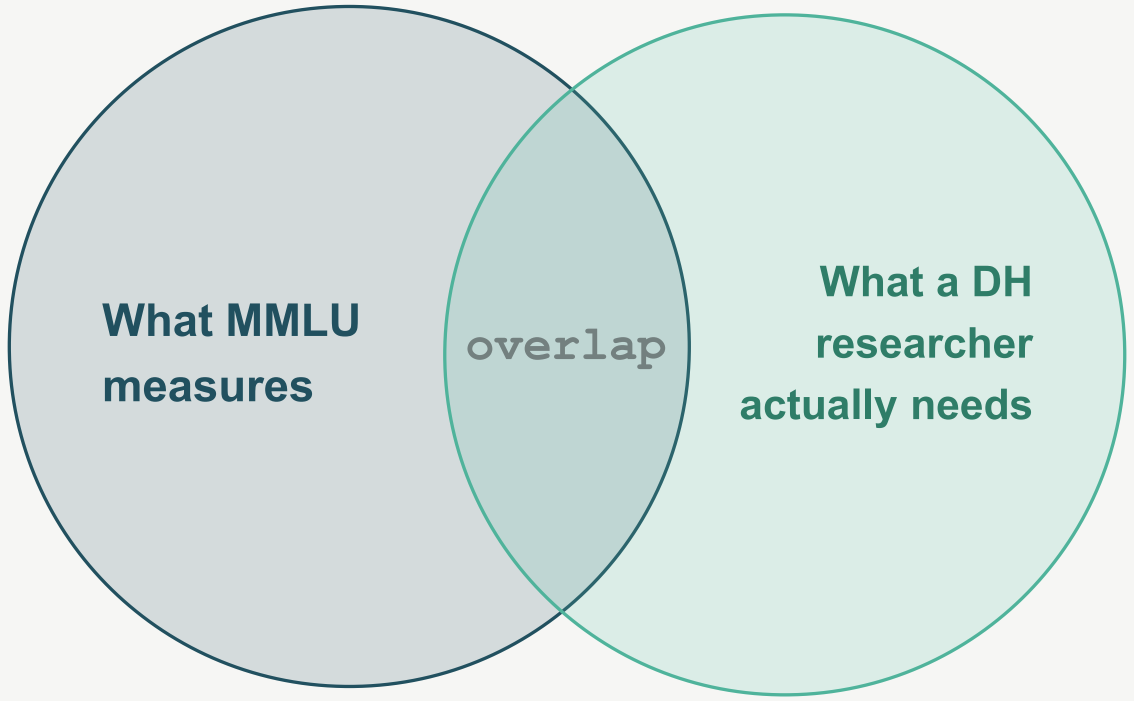


top 10 models · all 88–92%

A 2-point difference here tells you almost nothing.

The construct validity problem

Does the test measure the ability it claims to? A model can ace a reasoning leaderboard and still be poor at *your* job. Correlation is not identity.



Leaderboards: what they show and what they hide

01 Aggregation hides per-task variation.

02 Human preference ≠ accuracy — people prefer long, confident, fluent answers, even when wrong.

03 Leaderboards reward optimisation toward the leaderboard.

04 Tier labels (`apex/nexus/lite/nano`) sit on top of the rankings and obscure the real models.

Decoding the ARC platform

1. The "size" column is misleading — cost tracks **active** , not total parameters.
2. Quantisation (`INT4/FP4/Q4`) trades a little quality for speed and memory.
3. The **context window** is the constraint that matters for document tasks.

`arc:nexus`

size	`35B total · ~3B active ← pay for this`
quantisation	`Q4_K_M ≈ 4-bit`
context	`262,144 ← the real limit`

The prompt is part of the test

Same model, same record, two phrasings → clean output, a ramble, or a hallucinated date.

`prompt A — "Extract the date as YYYY."`

```
1481
```

clean · parseable

`prompt B — "When was this made?"`

```
"Probably the late 15th century, around 1490…"
```

prose · a guessed year

When you read "X beats Y," ask: *with what prompt, how many runs, what temperature?*

Five questions to ask any benchmark

01 What task format? Multiple choice ≠ open generation.

02 Could training data include the questions? Contamination.

03 Are the top models clustered? Saturation.

04 Does it measure what I actually need? Construct validity.

05 Who made it, and who benefits from the ranking? Incentives.

Roll your own

If you can't trust the leaderboard for your task, build your own.

If you can't trust the leaderboard for your task, roll your own.

https://nfcs-poc.sites.er.kcl.ac.uk

The evaluation design loop

1. **Define the task** →
2. **Create a gold standard** →
3. **Choose a metric** →
4. **Run the models** →
5. **Analyse failures**

↻ `loop back` *What did this reveal about the task definition?*

What is a gold standard?

Human-made correct answers, made *before* seeing model output. Good ones are representative, consistent (measure annotator agreement), and documented.

record_0427 · hand-annotated

"A small bronze fibula, decorated, of probable Late Iron Age date, recovered near East Yorkshire."

- object: bronze fibula
- period: Late Iron Age
- place: East Yorkshire

Even 10–50 examples reveal systematic patterns. Small and honest beats large and sloppy.

Metrics: precision, recall, F1

Precision
Of what it gave, how much was right.

Recall
Of what it should have found, how much it got.

F1
The balance of the two.

$$F_1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

```
gold     bronze fibula · Late Iron Age · East Yorkshire
model    fibula · Iron Age · Yorkshire
```

Is partial credit appropriate? The metric encodes a judgement about what matters.

Scoring the unscorable

No single right answer? Score against 3–4 observable, specific criteria — not "is it good?" but "does it identify the text's claimed origin?"

Criterion	Score (0–2)
Identifies the object type correctly	2
States the claimed period	1
Gives a defensible location	2
Avoids unsupported invention	0

The catch: the rubric carries its author's assumptions. **Two rubrics can disagree.**

Same model, different prompt, different answer

DIRECT / STRUCTURED

```
"Return object, period, place as JSON."
```

Tends to give **parseable** output.

ROLE-FRAMED

```
"You are a museum cataloguer. Describe this find."
```

Can do better on **ambiguous judgement**.

Test both models with the *same* prompts — or you're comparing model+prompt combinations, not models.

Temperature and repeatability

Models are non-deterministic by default. Use `temperature 0` (and ideally several runs) so differences come from the model, not the dice.

TEMPERATURE 1.0 — 3 runs

```
"c. 1490, East Yorkshire"
"Late Iron Age, N. England"
"~50 BCE, Yorkshire region"
```

varies — the dice are talking

TEMPERATURE 0 — 3 runs

```
"Late Iron Age, East Yorkshire"
"Late Iron Age, East Yorkshire"
"Late Iron Age, East Yorkshire"
```

stable — differences are the model

Letting a model do the scoring

A strong model can score outputs against your rubric — it scales, but it imports the judge's biases (it prefers long, confident, self-similar answers).

> A labour-saver under human spot-checking — never an oracle.